

Forecasting California Electricity Demand Using Weather Features and Machine Learning

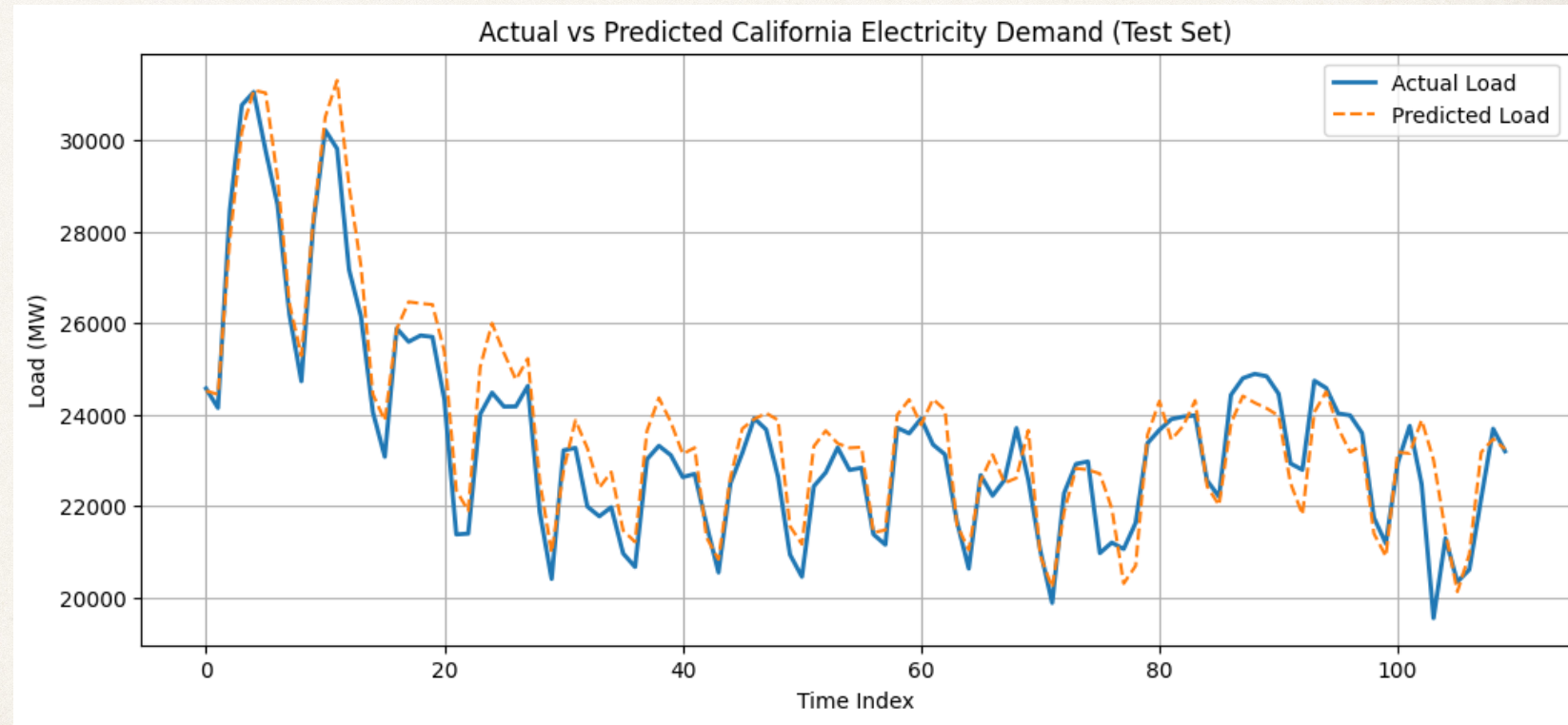
Jeremy Ferrara

[github repo](#) • [app](#)



Project Results at a Glance

- ❖ Built and deployed a live [4-day forecasting app](#)
- ❖ Final model:
Linear Regression
- ❖ Test $R^2 = 0.887$
- ❖ Average Forecast Error ≈ 597 MW



Forecasting App

❖ [app link](#)

❖ Live weather and CAISO demand data

❖ Recursive 4-day demand forecasts

❖ Built on final linear regression model

⚡ California Electricity Demand Forecast

This app estimates California's average electricity demand for the next four days.

The forecast is generated using a linear regression machine learning model trained on historical California electricity demand, weather patterns across five major California cities, and calendar-related features.

Daily forecast

Each card shows the model's predicted average electricity demand for that day. The weather shown below each prediction uses forecasts from Los Angeles, San Francisco, San Diego, San Jose, and Fresno.

Tue, May 19

Predicted average demand

22,209 MW

↑ 2,882 MW

Weather by city

☀ Los Angeles: Clear Sky

☀ San Francisco: Clear Sky

☁ San Diego: Broken Clouds

☀ San Jose: Clear Sky

☀ Fresno: Clear Sky

Statewide average temperature: 67.4°F

Total precipitation estimate: 0.0 mm

Wed, May 20

Predicted average demand

24,160 MW

↑ 1,952 MW

Weather by city

☀ Los Angeles: Clear Sky

☁ San Francisco: Scattered Clouds

☀ San Diego: Clear Sky

☁ San Jose: Few Clouds

☀ Fresno: Clear Sky

Statewide average temperature: 70.6°F

Total precipitation estimate: 0.0 mm

Thu, May 21

Predicted average demand

24,896 MW

↑ 735 MW

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Statewide average temperature: 69.7°F

Total precipitation estimate: 0.0 mm

Fri, May 22

Predicted average demand

25,308 MW

↑ 413 MW

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☁ San Francisco: Few Clouds

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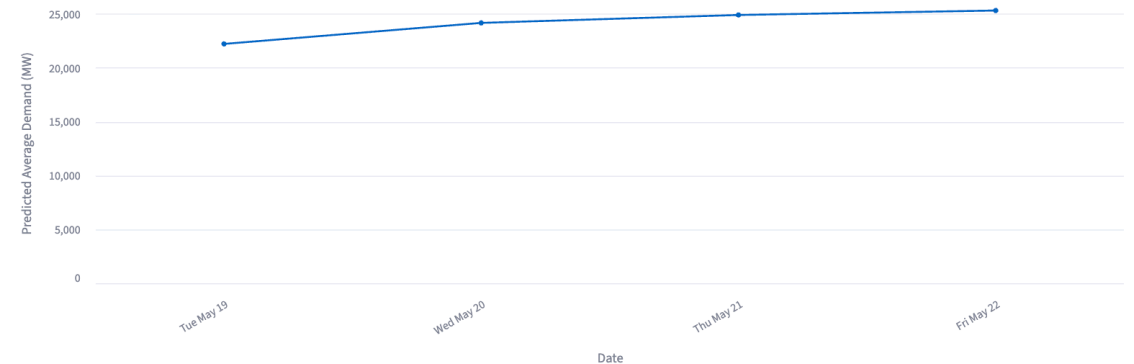
☀ Fresno: Clear Sky

Statewide average temperature: 69.4°F

Total precipitation estimate: 0.0 mm

Forecast chart

4-day forecast of predicted average California electricity demand.



Why Forecast Electricity Demand?

- ❖ Forecasting supports:
 - ❖ Grid stability
 - ❖ Energy planning
 - ❖ Cost management
 - ❖ Renewable energy integration
- ❖ Heat waves create major statewide demand spikes

Project Goal

- ❖ Build an interpretable machine learning system capable of forecasting short-term California electricity demand using weather and calendar-based features
- ❖ Deploy the model into a live forecasting app

The Data

- ❖ 2 years of weather data: **Los Angeles, San Francisco, San Diego, San Jose, Fresno.**
 - ❖ [Meteostat API](#)
- ❖ Electricity demand data:
 - ❖ [CAISO](#) (California Independent System Operator)
—> [GridStatus API](#)
- ❖ Live weather forecast: [OpenWeatherMap API](#)

```
start = datetime(2024, 1, 1)
end = datetime(2025, 12, 31, 23, 59)

points = {
    "la": ms.Point(34.0522, -118.2437),
    "sf": ms.Point(37.7749, -122.4194),
    "sd": ms.Point(32.7157, -117.1611),
    "sj": ms.Point(37.3382, -121.8863),
    "fresno": ms.Point(36.7378, -119.7871)
}

weather_dfs = {}

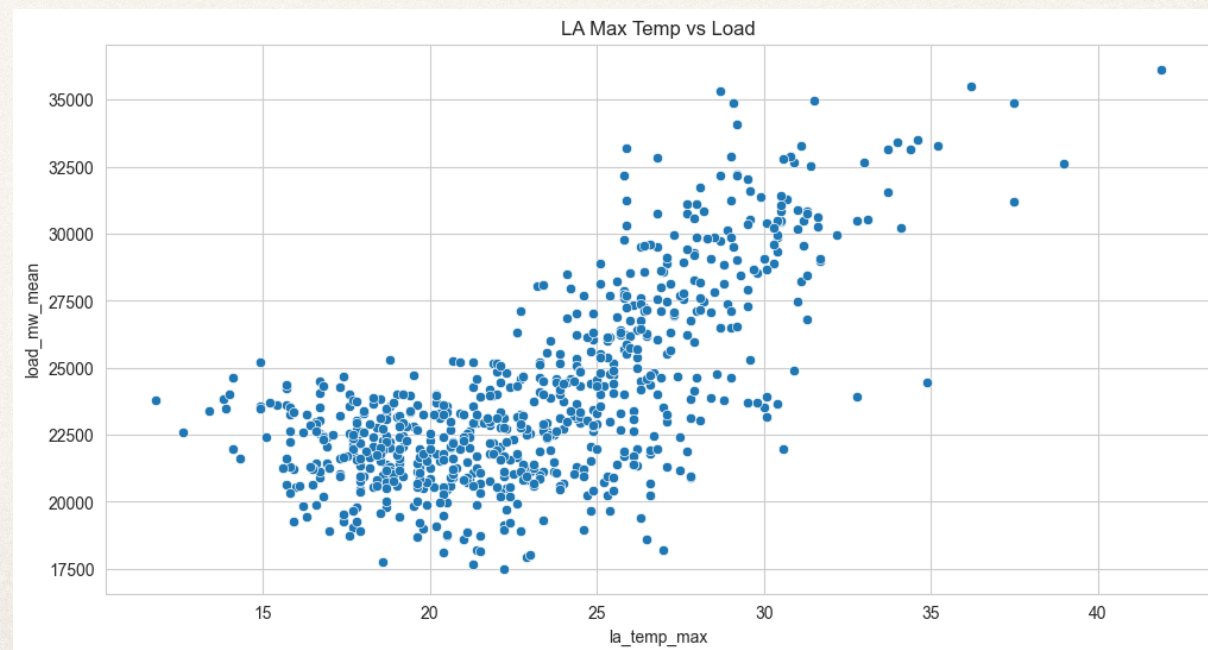
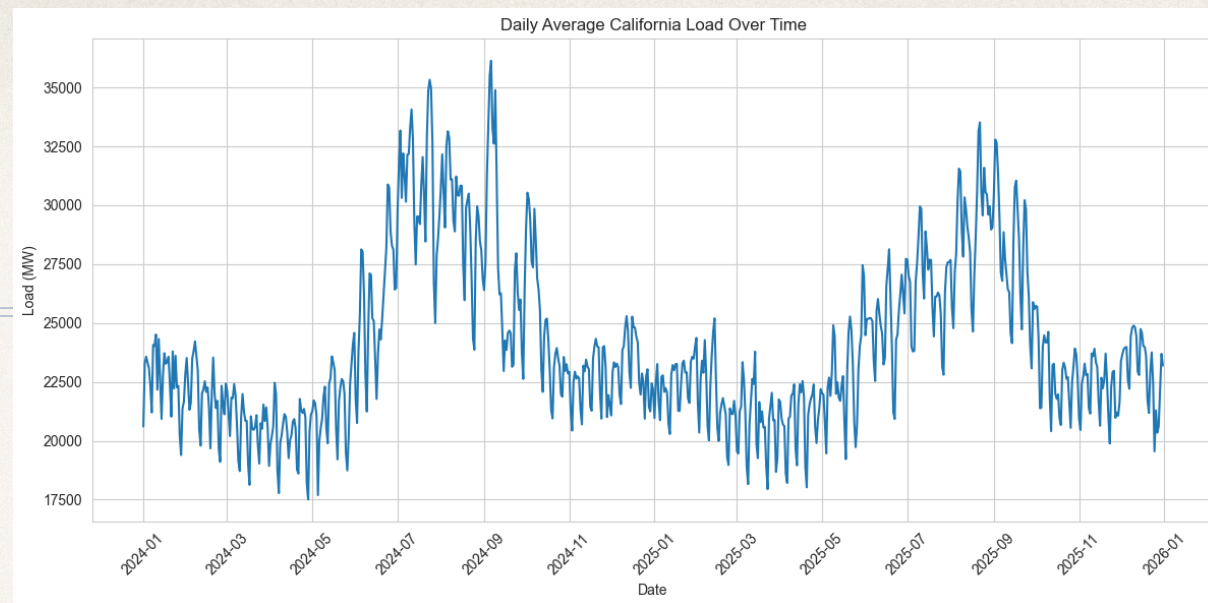
for name, point in points.items():
    stations = ms.stations.nearby(point, limit=4)
    ts = ms.hourly(stations, start, end)
    df = ms.interpolate(ts, point).fetch().reset_index()
    df = df.rename(columns={"time": "datetime"})
    weather_dfs[name] = df

    print(name, df.shape)
    print(df.head(2))
```

```
la (17544, 10)
  datetime temp rhum prcp wdir wspd wpgt pres cldc coco
0 2024-01-01 00:00:00 15.9 68 0.0 117 8.7 <NA> 1017.2 8 3
1 2024-01-01 01:00:00 14.2 74 0.0 98 7.8 <NA> 1018.8 7 3
sf (17544, 10)
  datetime temp rhum prcp wdir wspd wpgt pres cldc coco
0 2024-01-01 00:00:00 13.4 82 0.0 360 7.2 <NA> 1018.2 8 4
1 2024-01-01 01:00:00 13.7 79 0.0 360 7.8 <NA> 1018.1 8 3
sd (17544, 10)
  datetime temp rhum prcp wdir wspd wpgt pres cldc coco
0 2024-01-01 00:00:00 16.1 75 0.0 0 0.0 <NA> 1017.7 6 3
1 2024-01-01 01:00:00 14.4 84 0.0 290 7.6 <NA> 1017.8 5 3
sj (17544, 10)
  datetime temp rhum prcp wdir wspd wpgt pres cldc coco
0 2024-01-01 00:00:00 13.0 77 0.0 320 17.0 <NA> 1018.0 8 3
1 2024-01-01 01:00:00 13.3 77 0.0 320 16.6 <NA> 1017.7 8 3
fresno (17544, 10)
  datetime temp rhum prcp wdir wspd wpgt pres cldc coco
0 2024-01-01 00:00:00 14.0 72 0.0 360 7.0 <NA> 1019.0 8 3
1 2024-01-01 01:00:00 13.0 77 0.0 40 7.0 <NA> 1018.0 8 3
```


EDA

- ❖ Strong seasonal behavior
- ❖ Temperature most dominant predictor



Feature Engineering

- ❖ **Lag Feature**

- ❖ Previous day load (lag1_load)

- ❖ **Aggregated Weather Features**

- ❖ Statewide averages / max / min temperature
 - ❖ Humidity
 - ❖ Wind speed
 - ❖ Precipitation

- ❖ **Nonlinear Features**

- ❖ Cooling degree
 - ❖ Heating degree
 - ❖ Squared temperature features
 - ❖ Hot day indicators

- ❖ **Calendar Features**

- ❖ Weekend flags
 - ❖ Seasonal indicators
 - ❖ Cyclical encodings

Modeling Approach

❖ **Models Tested:**

- ❖ Linear Regression
- ❖ Ridge Regression
- ❖ Random Forest
- ❖ Gradient Boosting

❖ **Evaluation Metrics**

- ❖ MAE
- ❖ RMSE
- ❖ R^2

❖ **Train/Validation/Test split**

- ❖ 70/15/15

Model Comparison and Final Selection

- ❖ **Linear Regression**
outperformed more complex models:

- ❖ **Validation R^2 : 0.923**

- ❖ **Test R^2 : 0.887**

- ❖ Strong interpretability

- ❖ Lower complexity

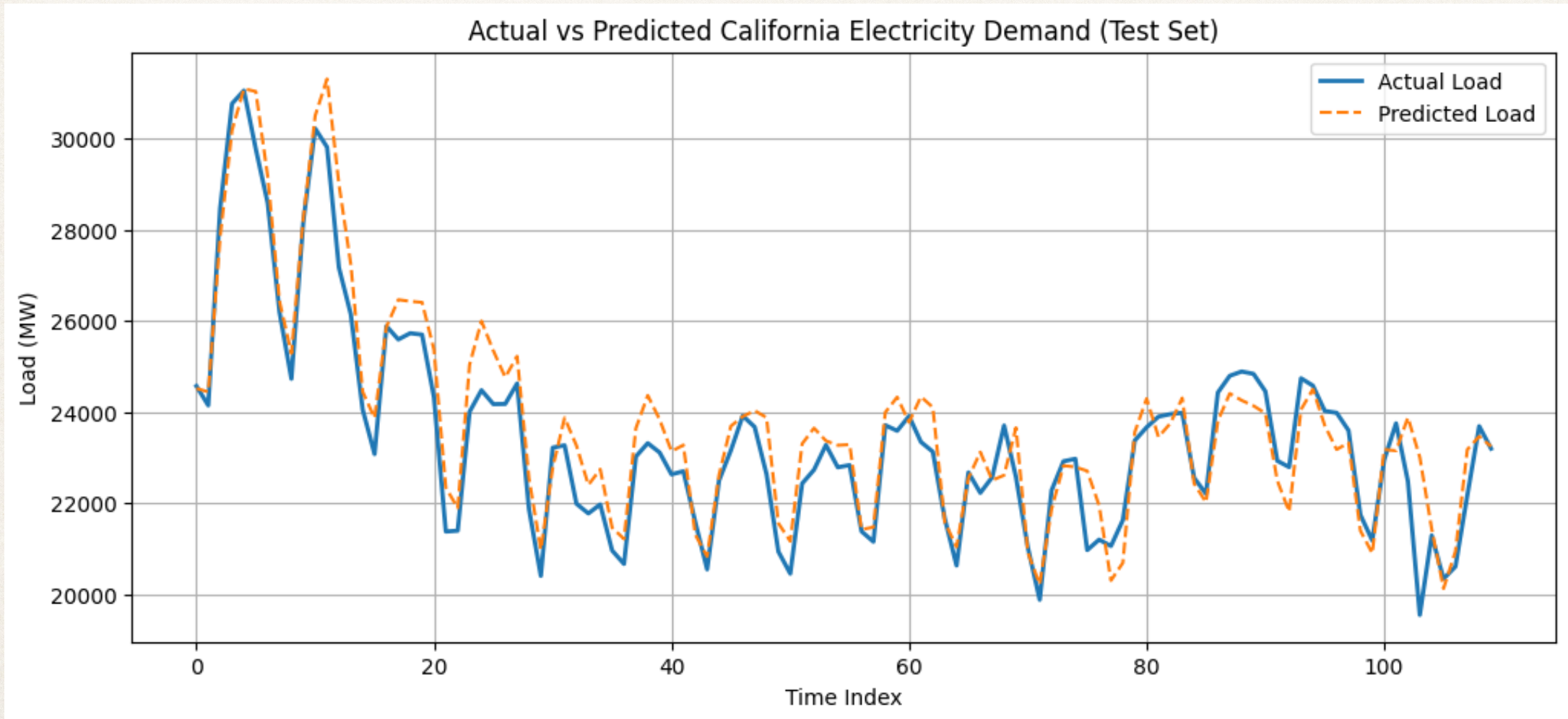
- ❖ Better generalization

	MAE	RMSE	R2	Model
0	564.934474	731.584943	0.923316	LinearRegression
1	640.426602	806.987540	0.906694	GradientBoosting
2	645.864222	844.789806	0.897748	Ridge
3	778.698579	941.074419	0.873111	RandomForest

Prediction Quality

❖ Average Forecast Error $\approx$ 597 MW

❖ Test $R^2 = 0.887$



Deployment

- ❖ Built and deployed a Streamlit [forecasting app](#)
- ❖ Live weather and CAISO demand data
- ❖ Recursive 4-day demand forecasts
- ❖ Recreates full training feature pipeline

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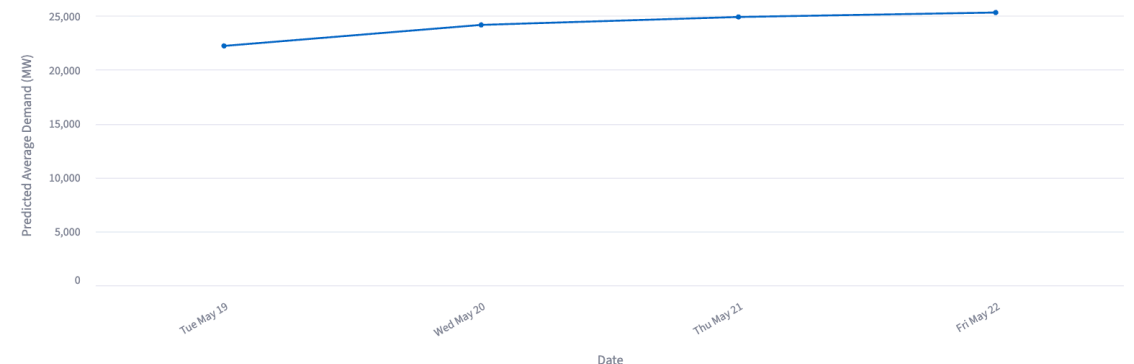
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Forecast chart

4-day forecast of predicted average California electricity demand.



Key Findings and Future Improvements

Key Findings:

- ❖ Temperature strongest demand driver
- ❖ Lag features significantly improved accuracy
- ❖ Feature engineering outperformed model complexity

Future Improvements:

- ❖ More historical training data
- ❖ Prediction intervals
- ❖ Max load prediction
- ❖ Advanced time-series forecasting models